CamillaFIR Full Case Study – From Measurement to Conclusion

This document demonstrates a complete CamillaFIR workflow example: Measurement → Filter Generation → Technical Report → Engineering Interpretation → Final Conclusion.

1. Measurement Stage

Room: Domestic listening room RT60 Wideband: ~0.24s Bass RT60 (63Hz): ~0.40s Crossover: 500 Hz & 2800 Hz (12 dB/oct) Correction Range: 10–200 Hz (magnitude) Max Boost: +3 dB Filter Type: Mixed Phase (split at 300 Hz)

2. Generated Dashboard

Initial Observations: - Strong LF room modes around 31 Hz and 113 Hz - Moderate reflections between 300–1500 Hz - Clean midrange response - Bass-first AI active

3. Summary Report Highlights

- Target Match: 94.4% (L) / 93.5% (R)
- RMS Error: 0.65–0.80 dB
- Acoustic Score: ~89/100
- Boost Peak: +3.00 dB (limit respected)
- Max Cut: ≈ -11 dB
- Phase Clamp: 54° → 45° protection applied
- XO GD impact: <0.6 ms (LOW)
- Hard clamp activated on 8 bins (boost protection)

4. Engineering Interpretation

Magnitude: The filter successfully reduced LF resonances without introducing excessive boost. Maximum boost limited to +3 dB ensures safe amplifier headroom. Phase: Phase correction was partially clamped, preventing unnatural time-domain behavior. Group Delay: Bass GD peaks correspond to room modes. Crossover GD impact is minimal and transient-safe. Smoothing (A-FDW): Adaptive smoothing (~1/5 octave average) prevents overfitting and ensures perceptual stability.

5. Risk Assessment

- No excessive boost (>3 dB)
- No clipping risk detected
- No unstable phase oscillation
- Bass decay remains room-limited (not EQ-fixable)
- Hard clamp indicates safety systems functioning correctly

6. Final Conclusion

The generated filter is technically sound and acoustically balanced. ✔ Excellent target match (>90%) ✔ Safe boost limits respected ✔ Controlled temporal behavior ✔ No aggressive HF overcorrection Remaining bass decay is a room property and would require physical treatment rather than additional EQ. Overall evaluation: Engineering-grade safe correction with high tonal fidelity.